

Data Science: Introduction to Machine Learning and Data Analysis

Course Introduction

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Getting to know your background

Please join Wooclap to answer a short background question.

Go to: www.wooclap.com then enter the code: PFNGLBI

or scan the QR code below:



Meet the Lecturers

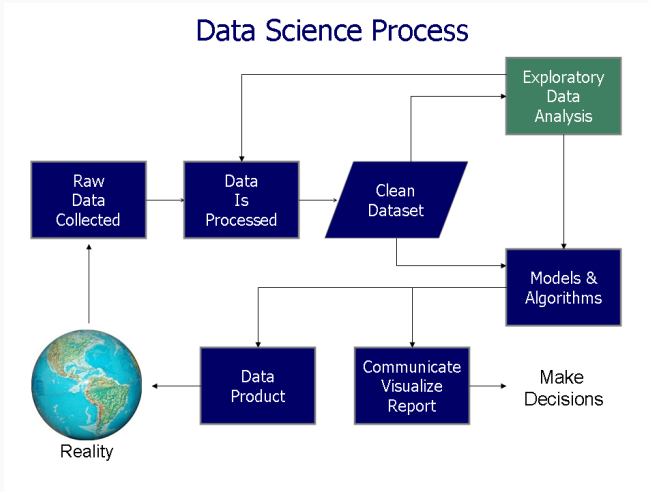


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Data science process

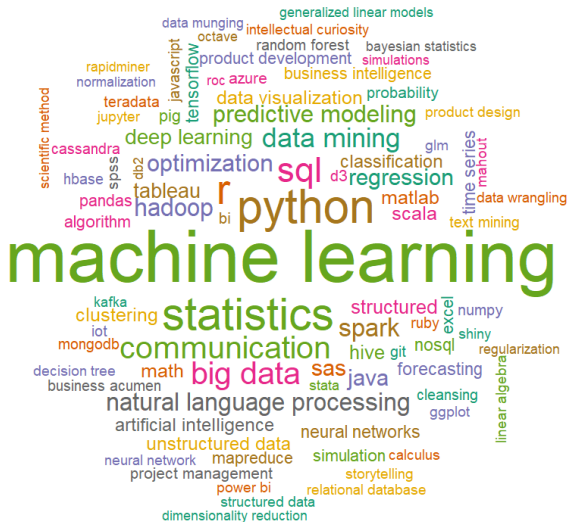


Machine Learning and Data Analysis

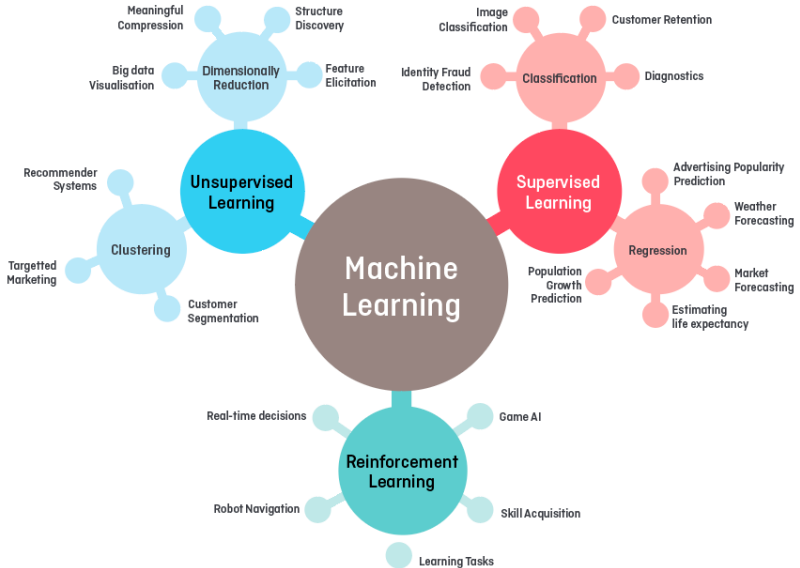
- **Data Analysis:** processing, cleaning, exploring, modeling, and visualizing data to **understand and interpret** it.
- **Machine Learning:** mainly using models and algorithms to learn patterns and then **predict or support decisions** on new data.

Machine learning is one part of data analysis and the broader data science process.

Data science word cloud



Models & Algorithms



Aim of the Course

By the end of this course, you should be able to:

- Explain the basic principles of data analysis and machine learning.
 - Visualize and explore data.
 - Distinguish between supervised and unsupervised learning.
 - Understand the basic ideas behind common learning algorithms.
 - Recognize and apply key statistical and machine-learning models.
 - Explain the bias–variance trade-off.
 - Apply and evaluate these methods using real-world data.
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- The course is **introductory and less technical**
 - The emphasis is on **understanding applications and practical use**



Introduction to Machine Learning and Data Analysis

Welcome to the website for the one-week Summer School course Introduction to Machine Learning and Data Analysis in R. This course offers a practical introduction to key methods in machine learning and data analysis using real datasets and hands-on examples. Participants are introduced to topics such as supervised and unsupervised learning, model validation, handling missing data, and presenting results in a transparent and reproducible way. The emphasis is on application rather than theory, with a focus on clear, interpretable methods that can be used in research practice. Here you can find the daily course materials for Monday through Friday.

Course Overview

Daily Schedule

Time	Activity
09:00–12:00	Lecture, lab, plenary discussion
12:00–13:00	Lunch
13:00–16:30	Lecture, lab, plenary discussion

Labs

Each session has a new lab. The links to the labs are available in the daily schedule above.

- HTML version of the lab for reading the exercises
- R Markdown version for working on the exercises yourself
- If you get stuck, use the code sections as support

Recommended Literature

- James, Witten, Hastie & Tibshirani (2021). *An Introduction to Statistical Learning with Applications in R*.
- Wickham & Grommund (2016). *R for Data Science: Visualize, Model, Transform, Tidy and Import Data*.
- ggplot2: *Elegant Graphics for Data Analysis*

Schedule and Course Materials

Monday

1A: Morning: Course Introduction and Data Visualization

- Lecture slides
- Exercise / lab

Afternoon: 1B: Bias-Variance Trade-Off

- Lecture slides
- Exercise / lab

Tuesday

Morning: Regression (Feature space expansions)

- Lecture slides
- Exercise / lab

Afternoon: Regression (Feature selection)

- Lecture slides
- Exercise / lab

Wednesday

Morning: Classification (Logistic regression and LDA)

- Lecture slides
- Exercise / lab

Afternoon: Classification (Trees and Support Vector Machine)

- Lecture slides
- Exercise / lab

Thursday

Morning: Unsupervised learning (Principal Component Analysis)

- Lecture slides
- Exercise / lab

Afternoon: Unsupervised learning (Clustering)

- Lecture slides
- Exercise / lab

Friday

Morning: Dealing with missing data in machine learning problems

- Lecture slides
- Exercise / lab

Afternoon: Discussion and Presentation

- Lecture slides
- Exercise / lab

Software

You need to have installed the following on your laptop, in this order:

1. R
2. RStudio

R packages

In this course, we use a wide range of R packages. It is highly recommended that you have administrator rights on your laptop, because this will allow you to install these packages when needed.

Practical Information

- Coordinator: Mahdi Shafiee Kamalabad
- Teachers: Mahdi Shafiee Kamalabad and Thom Volker
- Location: Room 2.07, Minnaert Building, Utrecht University
- Dates: 20–24 July 2026
- Contact: m.shafieekamalabad@uu.nl

Structure of the course

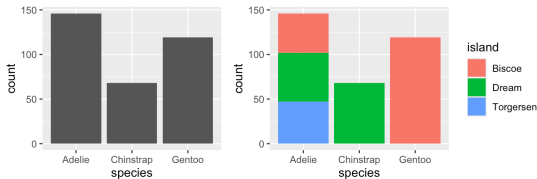
- Morning session (09:00-12:00)
 - lecture
 - lab
 - plenary discussion lab
- Lunch (12:00-13:00)
- Afternoon session (13:00-16:30)
 - lecture
 - lab
 - plenary discussion lab

The Labs

- The labs are presented as **HTML documents** with exercises
- The text of each exercise is visible
- The corresponding **R code** is hidden by default
- The code can be displayed by clicking the **SHOW** button

HIDE

```
p1 <- ggplot(peng, aes(species)) +  
  geom_bar()  
  
p2 <- ggplot(peng, aes(species, fill = island)) +  
  geom_bar()  
  
p1 | p2
```



c. An alternative for plotting two categorical variables is `geom_count()`, which displays the number of observations for each combination of categories. Plot the variables species and year (treating year as a factor) using this geom.

SHOW

- We encourage you to **first try to solve each exercise on your own**
- If you get stuck, you can consult the **code** to help you understand the solution
- You may also use **ChatGPT or similar AI tools** as additional support, but use them very critically. They are not always correct.
- If you use AI, use it to clarify concepts, debug errors, or check your understanding - not to replace your own thinking. Be critical. AI also makes mistakes...

